

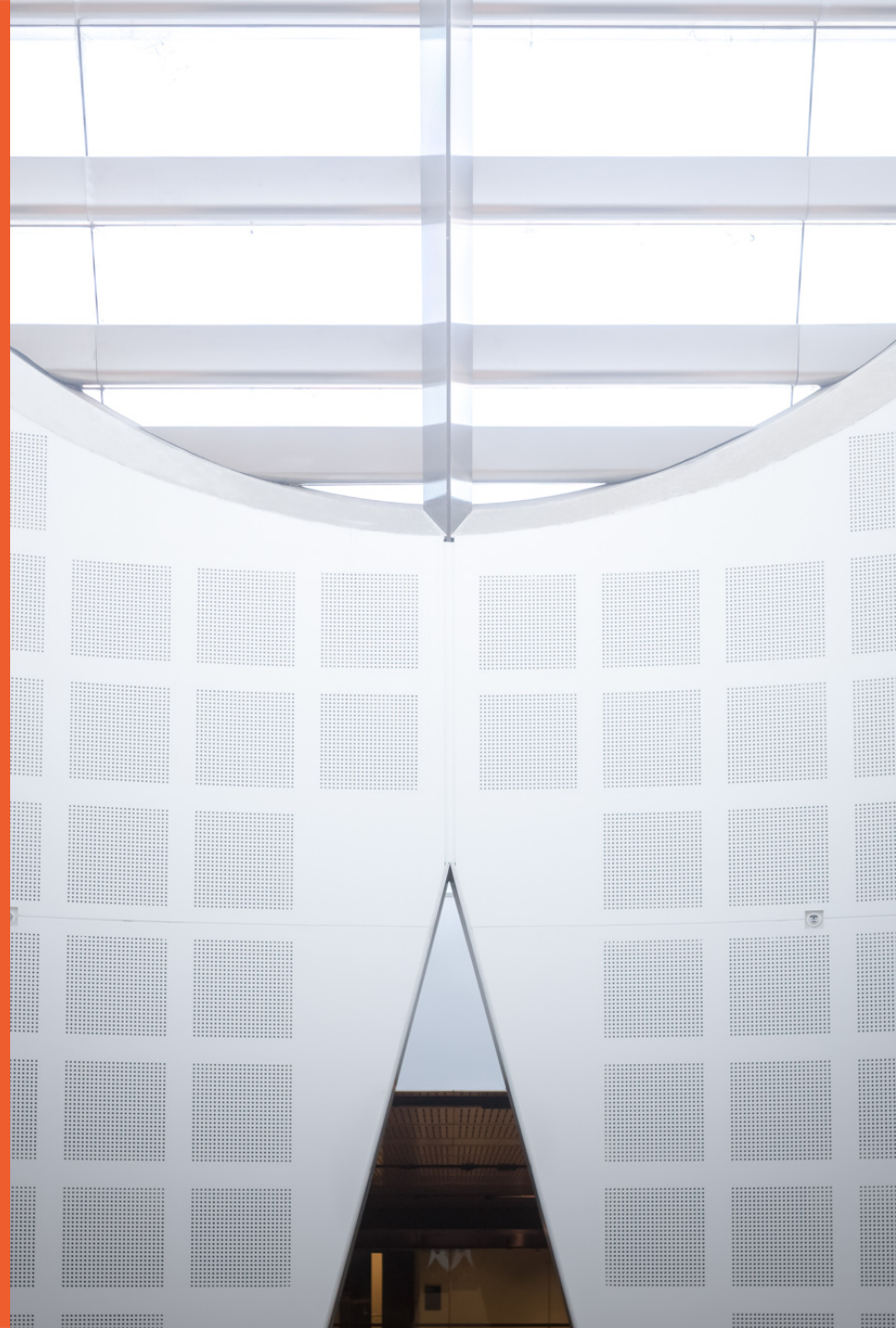
Experimental Design and Power Calculation @ Camden Campus

Presented by
Jim Matthews
Senior Consultant: Statistics
Sydney Informatics Hub
Core Research Facilities
The University of Sydney

2019



THE UNIVERSITY OF
SYDNEY



Outline

- The research question and experimental design
- Observational Studies and correlation vs causation
- Experimental validity
- Randomisation and bias
- Blinding and bias
- Blocking and confounding
- Fixed and random effects
- Replication, Experimental Units
- Exercise
- Statistical power and sample size calculation
- Software tools

The Research Question suggests the design options

What kind of research are you doing?

What is the best approach to answer your research question?

- Observational study
- Longitudinal observations – prospective or retrospective
- Field experiment
- Natural experiment
- Laboratory experiment
- Systematic review and meta-analysis
- Randomised controlled trial
- Clinical study
- Case-control study
- Other

Designed Experiments

We will mainly consider study types where you are designing an experiment.

The experimental conditions are designed and controlled.

- Identify and control the independent variables
- Randomisation to prevent bias and confounding
- Replication to control measurement uncertainty

Designed experiments can identify causal relationships between the independent (explanatory) variables and the dependent (response) variables.

Observational studies

Common in public health and epidemiology

Data is collected without experimental design (intervention)

Aim to discover *correlations* or *associations* between risk factors and health outcomes, eg air pollution and respiratory complaints

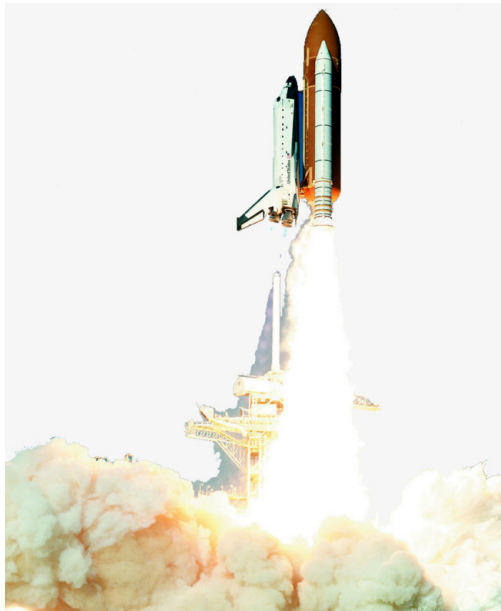
Are *typically* not capable of causal inference on their own

Often mis-reported by journalists

Correlation is not causation

Examples from tylervigen.com/spurious-correlations

US spending on science,
space, and technology



?



$r=0.998$

Suicides by hanging
and suffocation



Correlation is not causation

Examples from tylervigen.com/spurious-correlations

Films Nicolas Cage
appeared in



?



$r=0.666$

•Number of people who
drowned by falling into a
pool.



Correlation is not causation

Examples from tylervigen.com/spurious-correlations

Per capita consumption
of margarine



?



$r=0.993$

Divorce rate in
Maine, USA



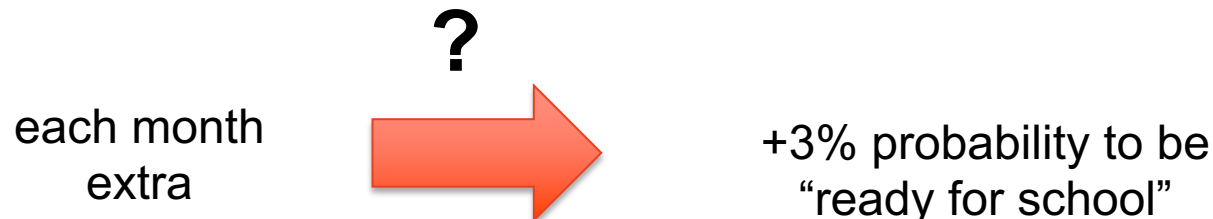
Correlation is not causation

'A gift of time': Children who start school later fare better, study finds – Sydney Morning Herald April 9, 2019 – 3:10am

Early Childhood
Research Quarterly

“A quarter of NSW children are starting school a year later than they are eligible, and **the delay is helping them fare better** in kindergarten than their younger peers, a landmark study has found.”

each month of maturity corresponded to an increase of about three per cent in the probability of scoring above the 25th percentile in five early development domains.



Epidemiology – weight of evidence

Bradford Hill Criteria

“The **Bradford Hill criteria**, otherwise known as **Hill's criteria for causation**, are a group of 9 principles, established in 1965 by the English epidemiologist Sir Austin Bradford Hill. They can be useful in establishing epidemiologic evidence of a causal relationship between a presumed cause and an observed effect and have been widely used in public health research. Their exact application and limits of the criteria continue to be debated.

Wikipedia “Bradford Hill Criteria” n.d.

Epidemiology – weight of evidence

Bradford Hill Criteria

1. **Strength** (effect size): A small association does not mean that there is not a causal effect, though the larger the association, the more likely that it is causal.
2. **Consistency** (reproducibility): Consistent findings observed by different persons in different places with different samples strengthens the likelihood of an effect.
3. **Specificity**: Causation is likely if there is a very specific population at a specific site and disease with no other likely explanation. The more specific an association between a factor and an effect is, the bigger the probability of a causal relationship.
4. **Temporality**: The effect has to occur after the cause (and if there is an expected delay between the cause and expected effect, then the effect must occur after that delay).
5. **Biological gradient**: Greater exposure should generally lead to greater incidence of the effect. However, in some cases, the mere presence of the factor can trigger the effect. In other cases, an inverse proportion is observed: greater exposure leads to lower incidence.
6. **Plausibility**: A plausible mechanism between cause and effect is helpful (but Hill noted that knowledge of the mechanism is limited by current knowledge).
7. **Coherence**: Coherence between epidemiological and laboratory findings increases the likelihood of an effect. However, Hill noted that "... lack of such [laboratory] evidence cannot nullify the epidemiological effect on associations".
8. **Experiment**: "Occasionally it is possible to appeal to experimental evidence".
9. **Analogy**: The effect of similar factors may be considered.

Experimental validity

- In developing the design of a study, internal validity and external validity need to be considered.
- **Internal validity** is the validity of the experiment for the particular sample chosen, and is mainly to do with proper control of other factors and elimination of unintended systematic effects.
- **External validity** is how generalizable the study is to a wider population and depends on the size and representativeness of the sample used.

External Experimental validity: Example

Study to evaluate the effect of a feed supplement on the growth of calves

Study Design:

- 2 groups: Std feed and std feed with supplement
- 20 calves in each group
- All calves are the same breed - Charolais
- All calves born in the same season
- All male
- All based on the one farm



What larger population does this sample represent?

What population might we wish to make inferences about?

External Experimental validity: Example

Study to evaluate the effect of a feed supplement on the growth of calves

Conclusions will be valid for Charolais bull calves raised under local conditions.

Findings of the study may not be valid for all calves:

- Other breeds
- Female (heifers)
- Grass fed?
- other

External Experimental validity: Example

Study to evaluate the effect of a feed supplement on the growth of calves

How do we improve the external validity?

- Start with determining what is the population of interest.
- Are we interested in male/female?
- Are we interested in other breeds?
- What else?

External Experimental validity: Example

Study to evaluate the effect of a feed supplement on the growth of calves

The study could now include:

- Gender: male, female
- Feed type: Grass and grain
- Breed: Charolais, Hereford
- Climate: temperate, arid

This will expand the external validity of the study to cover a much wider population, but potentially make the study more difficult to carry out.

Compromise is often necessary.

Experimental validity

- The main problem with experiments is external validity.
- Will the study be confirmed by other researchers and become accepted science?



Experimental validity

One way to deal with limited external validity is to combine several or many studies using a meta-analysis.

- Studies from different countries
- Studies from different varieties or breeds
- Studies from varied environments and geographies

Experimental Design – Randomisation

What is randomisation?

Random allocation of treatments to subjects

Why randomise?

So we can avoid:

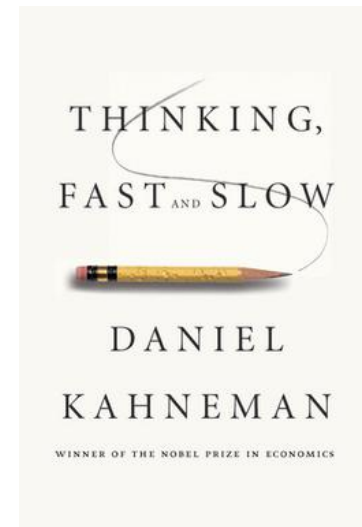
- Systematic bias – allocate all the drug treatments first, then the placebos
- Selection bias – subconsciously (or consciously!) choosing healthy patients for the treatment
- Unknown unknowns – potential confounding factors we don't even know exist.

Prevention of these biases prevents confounding and strengthens our causal inference

Biases and heuristics

Daniel Kahneman* describes the human psychology in terms of heuristics and biases. These include:

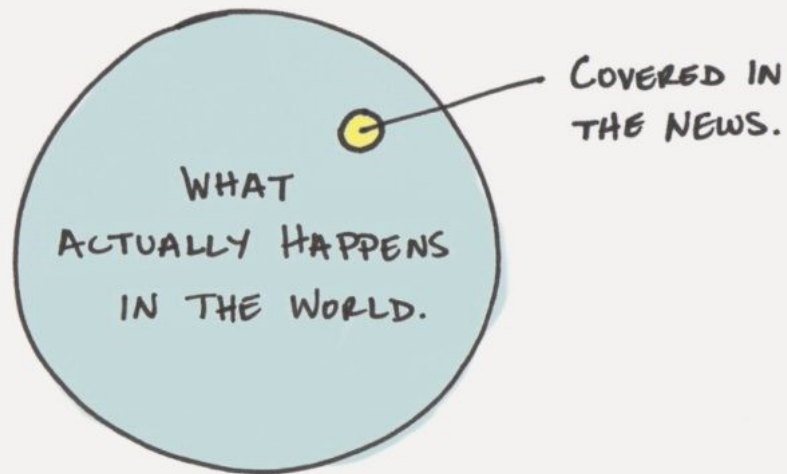
- Anchoring
- Availability
- Substitution
- Optimism and loss aversion
- Framing
- Sunk cost



* Daniel Kahneman wrote “Thinking, Fast and Slow”, 2011

Availability Heuristic

THE AVAILABILITY HEURISTIC



JamesClear.com

Experimental Design – CRD

Completely Randomised Design (unstructured design)

Example: Evaluate the effect of a feed supplement on the growth of calves

- Suppose that we have no information about the calves (subjects) that we might otherwise use. In this case we treat all subjects the same and use randomisation to eliminate bias.



Completely Randomised Design

- Suppose we have 20 subjects and 2 treatments (T and C)
- Allocate a code number to each subject from 1 to 20

TREATMENT CODE	T	T	T	T	T	T	T	T	T	C	C	C	C	C	C	C	C	C	C
-------------------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

- Generate a random sequence of numbers 1 to 20 (eg from Excel)
- In Excel use formulae: A1=rand(); B1=rank.avg(A1,A\$1:A\$20); copy down 20 rows.

random order	16	4	9	10	12	8	1	20	11	3	17	2	19	5	13	15	7	18	14	6
-----------------	----	---	---	----	----	---	---	----	----	---	----	---	----	---	----	----	---	----	----	---

- Another nifty random number sequence generator is at:
www.random.org/sequences
- Use a formal process to generate random allocation

Completely Randomised Design

- Now the “T” and “C” treatments can be applied to the subject codes using the random number sequence.

	T	T	T	T	T	T	T	T	T	T	C	C	C	C	C	C	C	C	C	C
random order	16	4	9	10	12	8	1	20	11	3	17	2	19	5	13	15	7	18	14	6

Feed treatment is allocated to these subject codes

Control treatment is allocated to these subject codes

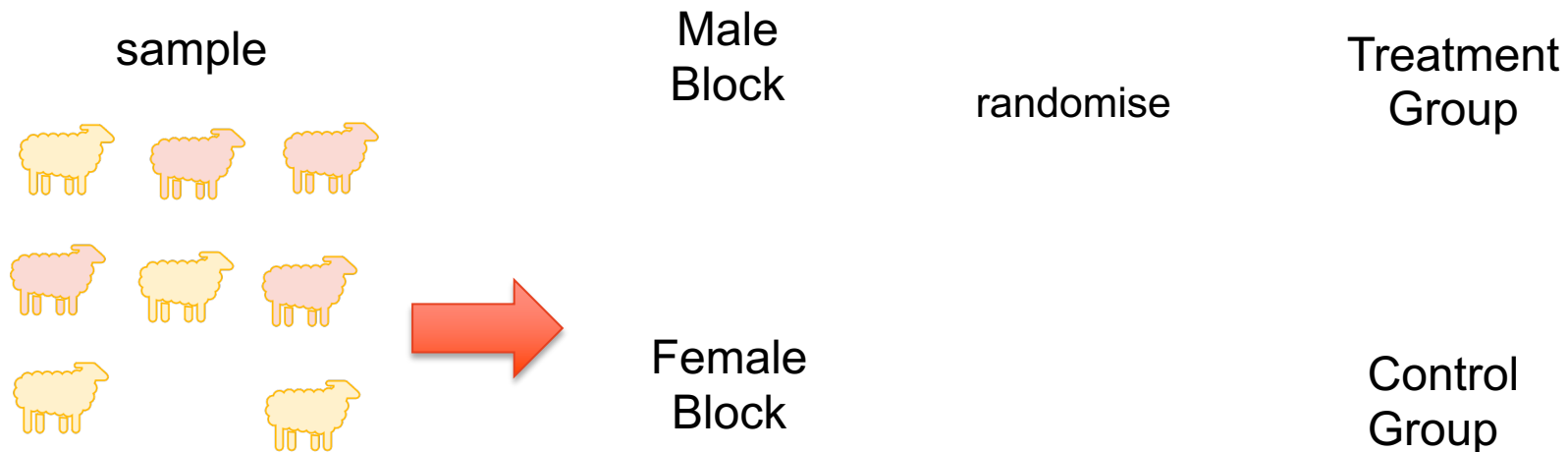
How will this experiment be analysed?

Experimental Design – Randomised Block Design

Randomised Block Design (RBD)

Example: Evaluate the effect of a feed supplement on the growth of calves

- Suppose now that we are able to source equal numbers of males and females.
- Use gender as a block variable and randomise within blocks.

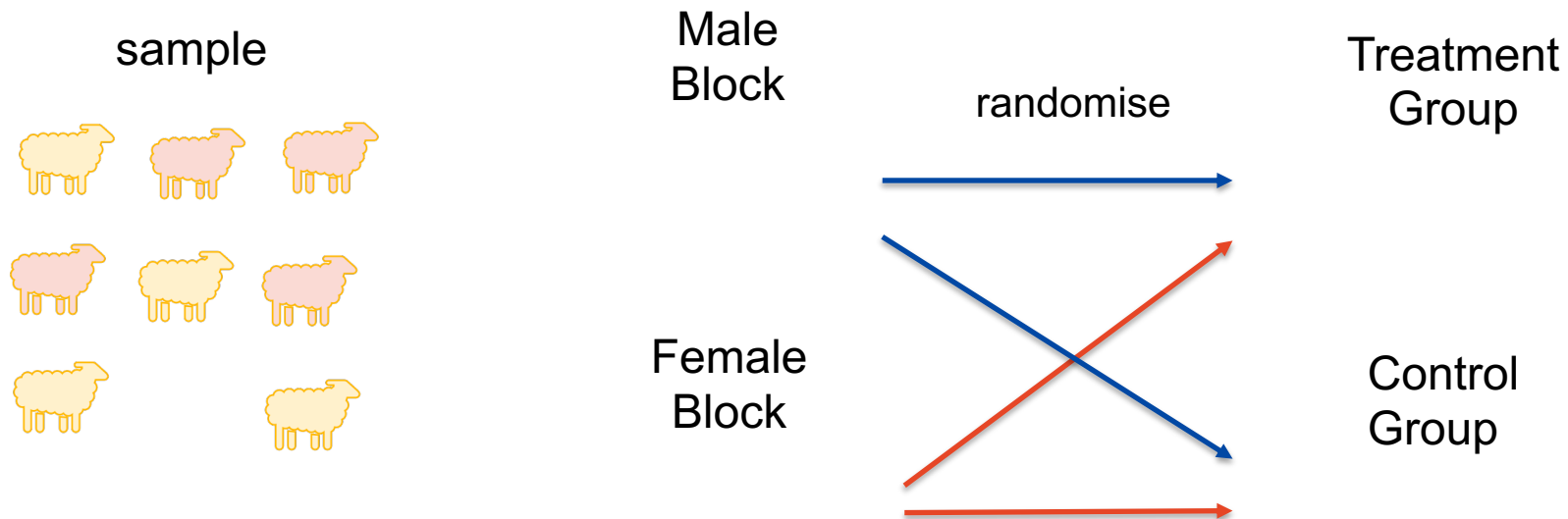


Experimental Design – Randomised Block Design

Randomised Block Design (RBD)

Example: Evaluate the effect of a feed supplement on the growth of calves

- Suppose now that we are able to source equal numbers of males and females.
- Use gender as a block variable and randomise within blocks.



Randomised Block Design

- Gender will be a block variable.
- Male Block

Treatment Code	T	T	T	T	T	C	C	C	C	C
Random order	10	8	2	1	4	9	3	7	6	5

- Female Block

Treatment Code	T	T	T	T	T	C	C	C	C	C
Random order	14	20	19	11	15	18	12	17	16	13

- The allocation is randomised within each (gender) block. Codes for M 1~10, codes for F 11~20.
- What would be the disadvantage of not blocking for gender in this case?
- How will this experiment be analysed?

Latin Square design

Used to create a balanced design with more than one blocking factor.

Example: Growing plants in a greenhouse.

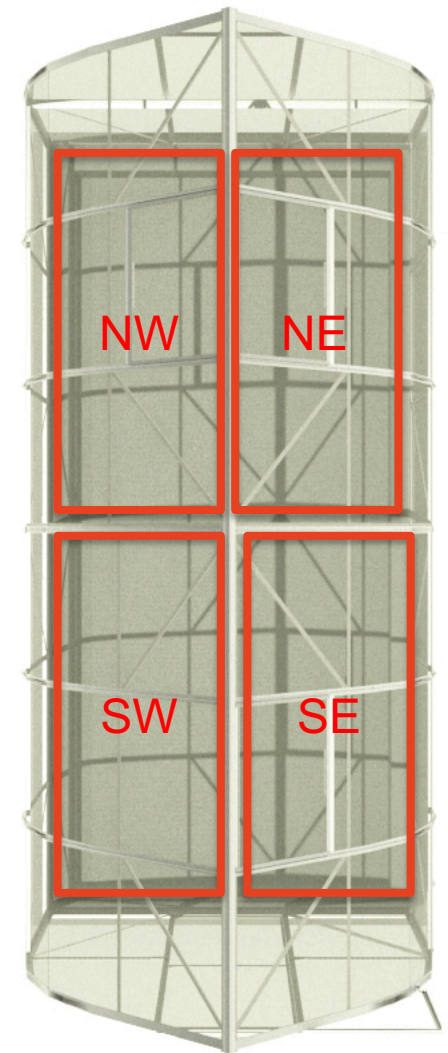
Treatments: A, B, C & D

Row block: shelf position 1,2,3,4

Column block: Corner position NE, NW, SE, SW

we can create a **4 x 4 Latin Square design**.

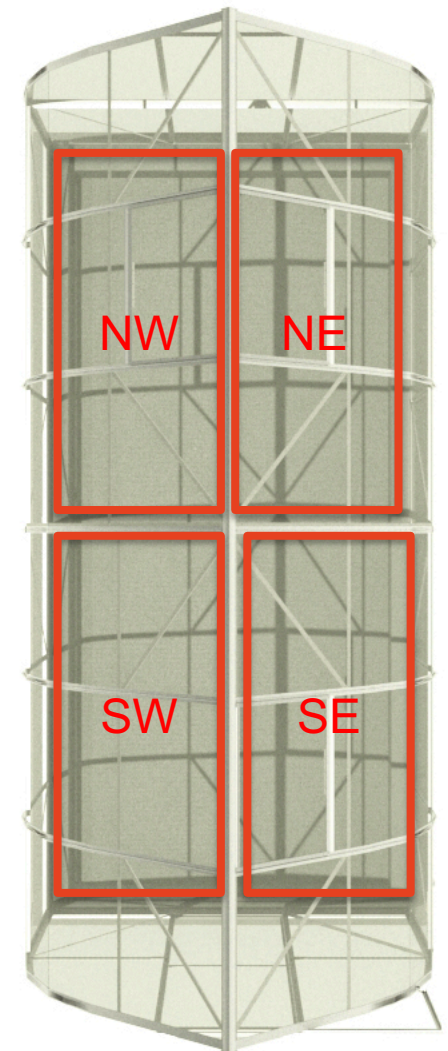
Each treatment occurs once per shelf position and once per corner position.



Latin Square design

Example: Growing plants in a greenhouse.

Column = corner position				
Row = shelf position	A	B	C	D
	D	A	B	C
	C	D	A	B
	B	C	D	A
	A	B	C	D



Balanced incomplete block design

Every pair of treatments occurs together in a block the same number of times.

Example: 4 treatments A, B, C & D

Block size is only 3 (not 4)

Block 1	A	B	C
Block 2	A	B	D
Block 3	A	C	D
Block 4	B	C	D

Don't forget, blocks could be batches, days, cycles, fields, etc

Factorial designs

Factorial experiments have 2 or more independent variables set at 2 or more fixed levels.

A common type is the two-level full factorial design

Example: Evaluate the effect of a feed supplement on the growth of calves

Factor A: Treatment [Supplement/std feed]

Factor B: Supplement quantity [low/high]

Factor C: Cattle breed [Charolais/Hereford]

This is a $2 \times 2 \times 2$ (or 2^3) factorial design

These designs allow the study of many factors for relatively few runs. Even fewer runs are required with fractional factorial designs. Works best with factors that can easily be controlled.

Experimental Design – Blinding

To avoid bias



Blind trials (or single blind) – the subject does not know if they are in the treatment or the placebo group

Double-Blind trials – Both the subject and the technician are not aware of the assigned treatment

Open trial – All the treatment information is known to the subject and technician/experimenter

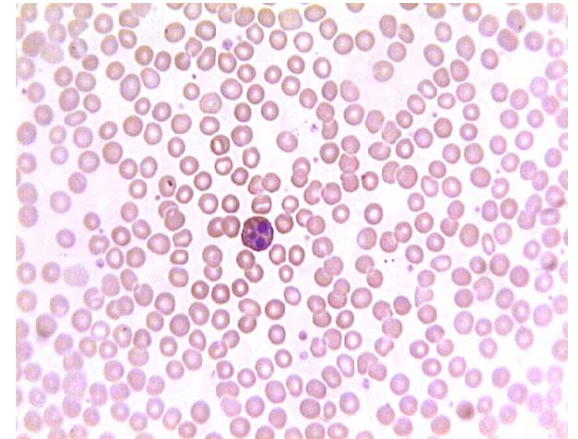
Blinding can reduce or eliminate confounding bias due to conscious or unconscious preferences or expectations.

Experimental Design – Blinding

Laboratory experiments can also benefit from blinding to prevent bias.

Example 1: Histology cell counts

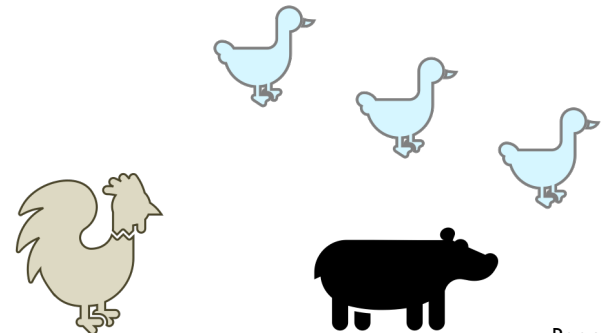
- Counting cells requires judgement (eg location sampling, recognising cell types)
- The technician should not know the identity of the specimens
- Use an ID code to anonymise the samples. Randomisation of processing order will also help



Example 2: Animal behaviour

- Many animals respond to the way they are handled
- The technician should (ideally) not know the identity of the animal's treatment group.

Can you use blinding in your research to guard against unconscious bias?



Fixed and Random Factors

Fixed Factors

- These are usually the independent variables chosen by the experimenter. They have defined levels or categories and we want to quantify the difference between them.

Random Factors

- These are usually incidental to the purpose of the experiment (such as blocking variables).
- The levels of the random factor may be chosen from a larger population of possible values of the variable.
- We don't need to quantify the size of the random effect, we are more interested in using it to partition variance and thereby reduce within group variance.

Fixed and Random Factors – calf feed example

Study to evaluate the effect of a feed supplement on the growth of calves

The study factors include:

- Gender: male, female
- Feed type: Grass and grain
- Breed: Charolais, Hereford
- Climate: temperate, arid

What might be fixed factors and what might be random factors?

Experimental and Observational units

Understanding the types of units will help you recognise design aspects such as sample size, repeated measures, technical replicates, and nested models.

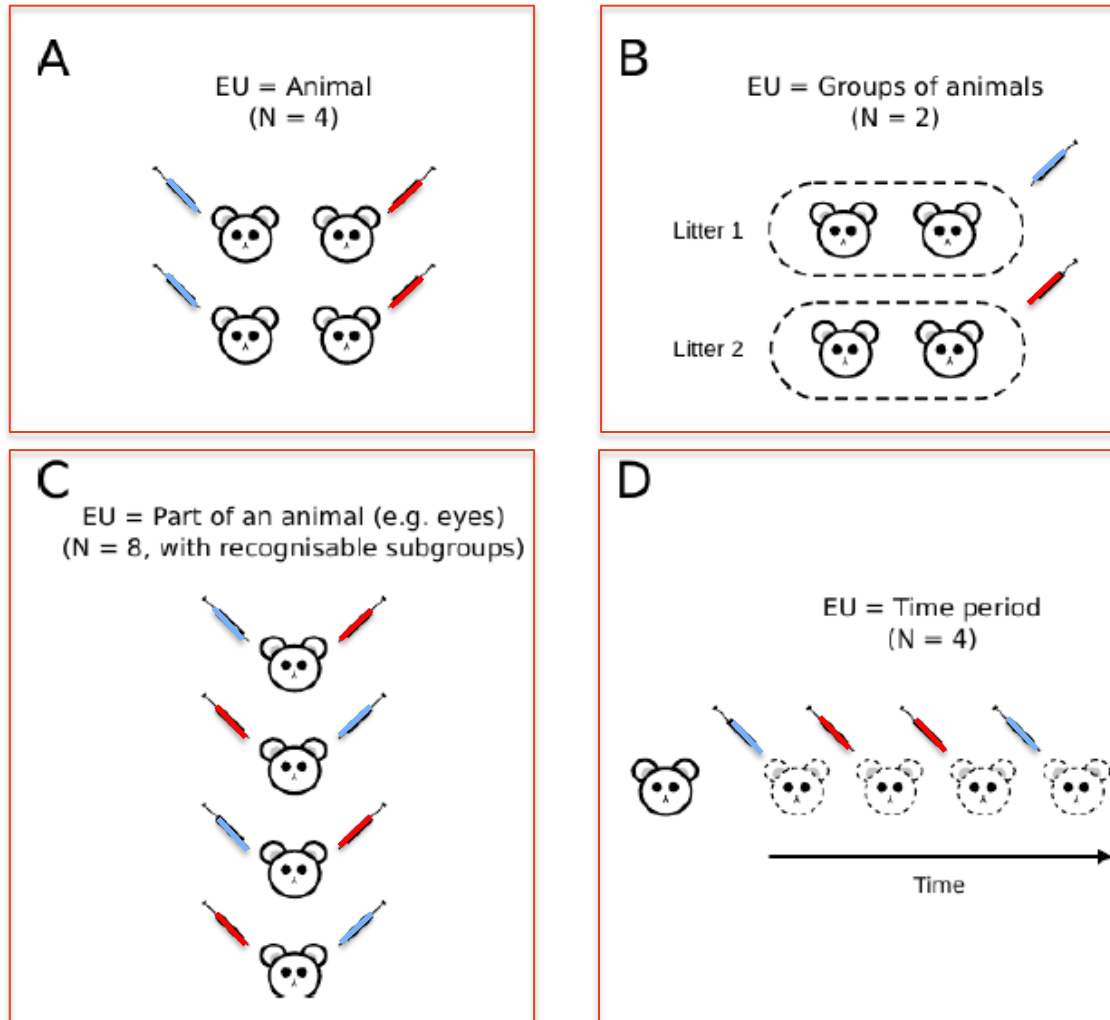
Types of Units (adapted from Lazic)

- Biological Unit – is the entity (animal/plant) about which inferences are made
- Experimental Unit – is the entity that is randomly and independently assigned to experimental conditions.
- Observational Unit – is the entity on which measurements are taken

What is the sample size n ?

n = number of experimental units

Experimental Unit leads to N



Experimental Unit leads to N

What about when the treatment is applied to the mother (mare, sow, ewe, etc) and the measurements are carried out on babies in the litter?

Identifying the BU, EU and OU can help.

Replication

True replication occurs when you have multiple independent measurements at the level of each experimental unit.

Pseudo-replication occurs when repeated measurements on a unit are not independent of each other.

Technical replication occurs when you take repeated measurements on a unit to increase the precision of that measurement (the measurements are averaged)

Repeated Measures

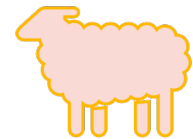
Repeated Measures Design (or within subjects design)

- Repeated Measures are not technical replicates when they represent another aspect of the same subject/sample, typically observations over time
- Repeated Measures are not independent observations
- There are specific statistical procedures to deal with RM's.

Scenario 1 – Sheep vaccine experiment

Research Question: Does the use of a new vaccine result in a different incidence of parasite infection compared to the standard treatment?

I would like 12 sheep in each group (total $n = 24$)
I have 12 sheep aged 1yr and 12 sheep aged 2yrs



Q: How should you allocate the treatments to the sheep?

Do vaccinate 6 of the younger sheep and 6 of the older sheep with each treatment.

Do not vaccinate 12 younger sheep with one vaccine and 12 older sheep with the standard vaccine treatment.

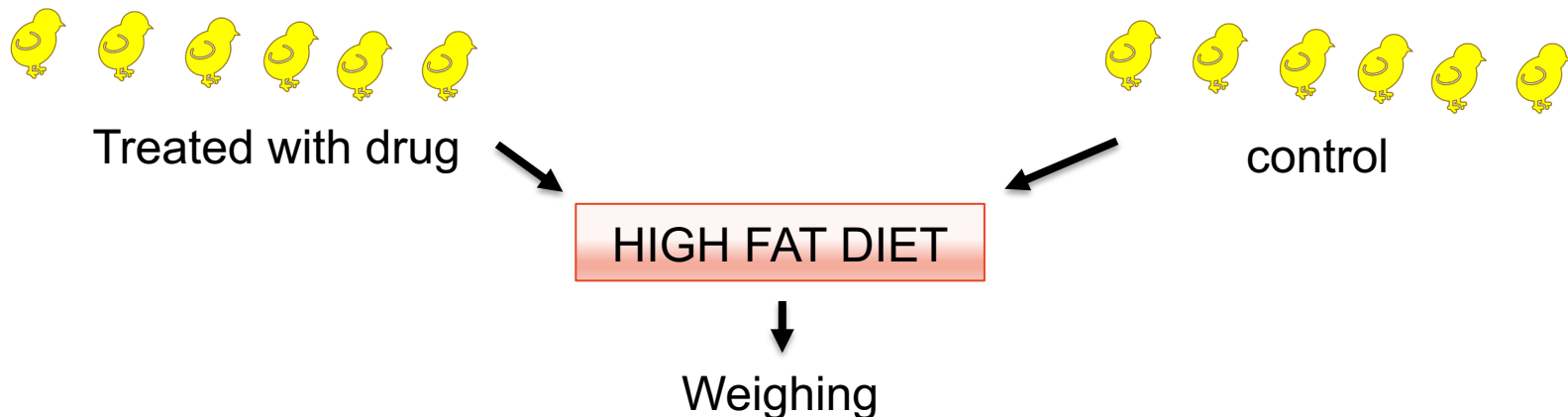
Avoid Confounding – occurs where the effect of two factors cannot be separately determined by the experiment.

Block Design – “sheep age” should be incorporated into the design as a blocking variable.

Scenario 2 – Chicken Drug & Diet experiment

Research Question: Groups of treated and untreated chickens are placed on a high fat diet. What is the effect on weight gain?

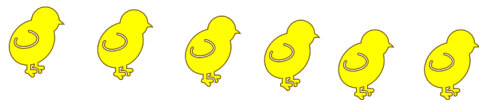
- What is the experimental design
- What are the dependent and independent variables
- What factors should be fixed in this experiment?
- What should be randomised?
- What factors should be blocked?



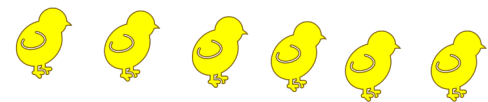
Scenario 2 – Chicken Drug & Diet experiment

What are:

Biological Units	
Experimental Units	
Observational Units	



Treated with drug



control



HIGH FAT DIET



Weighing

Worksheet - Chicken Drug & Diet experiment

Dependent Variables	Independent Variables	Design features	Blocking Factors	Randomisation & blinding

Worksheet - Chicken Drug & Diet experiment

Dependent Variables	Independent Variables	Design features	Blocking Factors	Randomisation & blinding
Weight gain	Drug treatment (y/n)	- Chick breed - High Fat Diet - Feeding routine - Feeding ad libitum? - Time of day for weighing - Housing – number of chicks per cage?	- Gender? - Chick age, batch, etc	- Drug treatment allocation - Order of handling - Order of weighing

Experimental Design – what statistical test will be used?

Worksheet – Your research

What does your research experimental design look like?

What are the biological, experimental and observational units?

Biological Units	
Experimental Units	
Observational Units	

Worksheet – Your research

Dependent Variables	Independent Variables	Design features	Blocking Factors	Randomisation & blinding

Experimental Design – what statistical test will be used?

Variation - blocking and randomisation - summary

1. Wherever possible, reduce or eliminate variation due to factors other than the independent variable.
 2. Sometimes undesirable variation cannot be avoided due to things beyond your control.
 3. Use blocking variables in your design to manage factors that are likely to cause variation.
 4. Use randomisation to prevent bias due to factors unknown or outside your control
 5. Use replication to improve precision of your estimated effects
- Point out the problems
 - Discuss the implications
 - Propose a way forward

What is the power of an experimental design?

The power to know...

Start with the hypothesis that you have generated, for example:
“The means of two groups are different”

In statistics, this is referred to as the alternative hypothesis H_1 .
Classically we test the veracity of the null hypothesis:

H_0 : There is no difference between the means of the two groups

A statistical test of the null hypothesis is always subject to uncertainty, or error. There are two main types of error.

Types of statistical error

Type I error

- Incorrectly rejecting the null hypothesis
- Also called false positive rate
- Referred to as the Significance level, designated by α
- The *convention* is to set the significance level to $\alpha = 0.05$

Type II error

- Incorrectly accepting the null hypothesis
- Also called the false negative rate
- Denoted by β
- Power is the complement of Type II error, denoted by $1 - \beta$
- We want Power to be as high as possible, typically $1 - \beta > 0.8$

Power calculation

How do we estimate the power of an experiment?

- It will depend on:
 - Sample size (more samples = more power)
 - Chosen significance level (typically $\alpha = 0.05$)
 - Minimum effect size (larger minimum effect = more power)
 - Variance within groups (larger variance = less power)
 - Experimental design and type of statistical hypothesis test

Sample size calculation

Often we need a sample size given a required minimum power

- Determine experiment type and statistical test
- Set α and $1 - \beta$
- Set the minimum effect size, standard deviation, proportions, as required by the design
- Calculate the minimum sample size

Sample size calculation

Setting values of parameters

- Typically choose $\alpha = 0.05$
- Typically choose $1 - \beta = 0.8$ (or higher)
- Decide on a minimum effect size of interest. This should be based on the minimum effect size that is of biological or clinical interest.
- Estimate the “variance” of the experiment. This will depend on the type of study. In some cases it will be difficult to estimate.

Estimation of study variance

Within study variance is the big unknown in this calculation

How to estimate it?

- Estimate standard deviation (or proportions) from previous experiments?
- Seek expert knowledge?
- If no idea, may be best to do pilot study

Power calculation software

Free calculation software

- R Studio: You can install a package, eg “pwr”
- Online calculators such as www.powerandsamplesize.com
- G*Power is a dedicated program

Proprietary \$\$ software

- Packages such as SPSS and SAS include a calculator
- GraphPad have “StatMate” separate to Prism
- PASS by NCSS dedicated software esp. for medical research

Power calculation software

G*Power

- Download from website:
- <http://www.psychologie.hhu.de/arbeitsgruppen/allgemeine-psychologie-und-arbeitspsychologie/gpower.html>
- Program has a simple user interface

The screenshot displays the G*Power 3.1 software interface. The window title is 'G*Power 3.1'. The 'Central and noncentral distributions' tab is selected. The 'Test family' is set to 't tests' and the 'Statistical test' is 'Means: Difference between two independent means (two groups)'. The 'Type of power analysis' is 'A priori: Compute required sample size - given α , power, and effect size'. The 'Input parameters' section includes 'Tail(s)' set to 'One', 'Effect size d' set to '0.5', ' α err prob' set to '0.05', 'Power (1- β err prob)' set to '0.95', and 'Allocation ratio N2/N1' set to '1'. The 'Output parameters' section lists 'Noncentrality parameter δ ', 'Critical t', 'Df', 'Sample size group 1', 'Sample size group 2', 'Total sample size', and 'Actual power', all with question marks. The 'Calculate' button is visible at the bottom right.

Central and noncentral distributions Protocol of power analyses

Test family: t tests Statistical test: Means: Difference between two independent means (two groups)

Type of power analysis: A priori: Compute required sample size - given α , power, and effect size

Input parameters:

- Tail(s): One
- Effect size d: 0.5
- α err prob: 0.05
- Power (1- β err prob): 0.95
- Allocation ratio N2/N1: 1

Output parameters:

- Noncentrality parameter δ : ?
- Critical t: ?
- Df: ?
- Sample size group 1: ?
- Sample size group 2: ?
- Total sample size: ?
- Actual power: ?

Calculate Effect

Calculate and transfer to main window

Close effect size drawer

X-Y plot for a range of values Calculate

Other resources

Books on Experimental Design

- *"The Design of Experiments"* by Fisher, Ronald Aylmer, 1935.
- *"Experimental Design for Laboratory Biologists: Maximising Information and Improving Reproducibility"* by S.E. Lazic
- *"Statistics for Experimenters"* by Box, Hunter & Hunter

Books on Causality

- *"The Book of Why"* by Judea Pearl (interesting ideas on causality, confounding, approaches to data)

Books on Bias and Statistical thinking

- *"Thinking, Fast and Slow"* by Daniel Kahneman

End of Workshop

- Thank you for your interest and attention
- Questions and comments welcome

- **Jim Matthews** BEng MStat | Senior Consultant: Statistics
- The University of Sydney
- Sydney Informatics Hub | Core Research Facilities
- Sydney Informatics and Data Science Hub (J17) | 32 Queen St Chippendale | The University of Sydney | NSW | 2006
- +61 412 246 271
- Jim.Matthews@sydney.edu.au | sydney.edu.au

